



# Saksi

A confidential holder register for tokenized real-world assets. Every position witnessed, no position revealed.

Cleanverse Build: Trusted Assets Hackathon · RWA Track · August 2026

Pugar Huda Mantoro · [hudapugar@gmail.com](mailto:hudapugar@gmail.com)

# Tokenization made the holder register public. Institutions never agreed to that.

An institution that buys a tokenized note broadcasts its position size, its entry timing, and its counterparty to every competitor watching the chain.

## What leaks today

- Position size, per holder, per asset
- Entry and exit timing, to the block
- Counterparty graph across the whole book
- Concentration — who the anchor investor is

## What it costs

A public register is a standing disclosure obligation no regulator asked for and no allocator wants. So institutional RWA retreats to permissioned ledgers, where the liquidity is not — or the issuance never happens at all.

Of the projects building on Cleanverse infrastructure this cycle, every one publishes its holder positions in the clear.

# The obvious fix breaks the thing that made it legal.

## Public register

**Compliant, not confidential.**

The issuer can prove eligibility, enforce transfer restrictions, and answer a regulator — but every holder trades in the open.

## Privacy pool

**Confidential, not compliant.**

Positions are hidden — and so is the register. The issuer can no longer prove who holds what, enforce a concentration cap, or find a holder whose credential was revoked.

## Saksi

**Confidential to the market, answerable to the issuer and the regulator.** Positions are shielded on-chain. Eligibility is proven inside the circuit. Compliance facts are proven on-chain without revealing the underlying positions.

# Private in the middle, accountable at the edges.

## ENTER

Holder deposits CVA. A zero-knowledge proof shows their key is in the CVI registry and absent from the sanctions set. CCP pre-checks; Travel Rule data attaches.

## HOLD & TRANSFER

Positions live as commitments. Amount and holder are hidden on-chain. Transfers re-prove eligibility every time — never once at onboarding.

## EXIT

Withdrawal releases CVA to a verified wallet only, through the same rule check, with attribution to the beneficial owner.

## Answerability, without disclosure

Four proof types, each verified by an on-chain contract: an exact figure, a threshold ("this holder is under the cap"), an aggregate ("total exposure to this jurisdiction is under the limit"), and a two-sided range. Each is bound to an audit request registered on-chain, so a proof cannot be recycled.

## Revocation as a live event

A revoked CVI enters the deny set. The holder's next proof fails non-membership and the position freezes — enforced by the verifier, not by an off-chain watcher running hours behind.

# Cleanverse is not a feature here. It is what the cryptography binds to.

| PRIMITIVE   | HOW SAKSI USES IT                                                                                                                                                                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CVI         | The membership set proven <b>inside the circuit</b> — not an off-chain KYC lookup before a transaction, but a constraint the proof cannot satisfy without a valid credential. The source key is pinned to the authenticated sender, so nobody enters as someone else. |
| CVA         | The only asset the pool accepts and the only thing it releases. The shielded leg moves A-Tokens strictly between verified wallets.                                                                                                                                    |
| CCP         | Pre-transaction rule check at both edges. If the check blocks, no proof is built and nothing moves.                                                                                                                                                                   |
| Travel Rule | Attribution rides every entry and exit, with look-through to the beneficial owner rather than to the pool contract.                                                                                                                                                   |
| Reporting   | An audit pack where each line is a claim <b>verified on-chain</b> : proof, verification hash, and the audit request it was bound to — not a spreadsheet asking to be trusted.                                                                                         |

Remove Cleanverse and there is no revocable, wallet-bound, bank-verified identity to prove membership in. The circuit has nothing to bite on and Saksi is an anonymous mixer — the exact thing this hackathon rules out of scope.

# Seven circuits, one pool, two consoles.

## Circuits

Circom / Groth16 / BN254. Compliance (membership + sanctions non-membership), shielded transfer, trustless tree update, and four selective-disclosure variants.

Exported to Solidity verifiers via snarkjs. BN254 has EVM precompiles, so verification is cheaper here than on the platform this code came from.

## Pool contract

Custodies CVA. Maintains the commitment set, the nullifier set, and the accepted-root history. Calls the verifiers on every state change and routes each edge transaction through CCP.

Deployment target: Monad. We probed all ten documented chains directly — Cleanverse's on-chain Validator answers on six of them, including the sponsor chain.

## Consoles

**Issuer** — register health, concentration and jurisdiction limits, revocation actions, audit-pack export.

**Regulator** — raise an audit request on-chain, receive the proof, verify it against the live contract.

## Honest scope

The zero-knowledge core is prior work, built and verified on Stellar testnet across eight contracts. The hackathon build is the port to EVM, the CVI and CVA gating, the Solidity pool, and the two consoles. Stated up front so the claim is checkable.

# Two moments, both on-chain, both live.

## One — prove without revealing

An issuer proves, on-chain, that no single holder exceeds the concentration cap and that aggregate exposure to a restricted jurisdiction stays under its limit.

**Not one position is disclosed.** The regulator's console verifies each proof against the live contract and prints the verification transaction hash.

## Two — revoke and watch it freeze

A holder's CVI is revoked. Their next transfer is attempted in front of the panel and the proof fails non-membership.

**The position freezes in the same block** — enforced by the verifier contract, with nothing off-chain in the loop.

Most compliance demos show a transaction succeeding. The interesting one is the transaction that **cannot** succeed.

# Confidentiality is the unbuilt half of the RWA stack.

**\$16T**

Projected value of tokenized assets by 2030, per the market context Cleanverse cites in its own launch materials.

**\$2.36T**

APAC crypto transaction volume in the year to June 2025, up from \$1.4T the year prior — the region Cleanverse's consortium sits in.

**0**

Projects in this hackathon's registered field offering a confidential holder register. Every one publishes positions in the clear.

## Who feels the pain

Issuers of tokenized notes, private credit, and funds who have a real concentration or jurisdiction limit to report against — and whose allocators will not accept a public book.

## Who pays

The issuer, not the holder. Saksi attaches to an existing CVA issuance rather than competing with it, so the buyer is already inside the Cleanverse network.



# A module, not a destination.

The disclosure layer is asset-agnostic. Any CVA issuer can attach a confidential register to an existing token without changing their issuance flow — which makes this infrastructure to be licensed, not a marketplace to be filled.

| LINE                  | BASIS                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------------|
| Register subscription | Per issued asset, per year. Covers the pool deployment, the verifier set, and the issuer console. |
| Settlement fee        | A thin basis-point charge on value entering the shielded register.                                |
| Audit pack            | Per regulator-facing report generated, where each line carries an on-chain verification hash.     |

**Position in the Cleanverse ecosystem**

Infrastructure Partner. Saksi does not issue, custody, or ramp. It makes an existing CVA issuance confidential without making it opaque — a capability the current stack does not offer, and one the consortium's own members are the natural first buyers of.

Pre-revenue. Pricing above is the proposed model, not observed willingness to pay.

# One issuer with a limit to report against.

| PHASE                       | WHAT HAPPENS                                                                                                                                                                                                                                             |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hackathon</b><br>Aug 8–9 | Circuits ported to Solidity verifiers — already done and gas-measured at 264k. Pool deployed on Monad and registered with Cleanverse's on-chain Validator against live CVI and CVA. Issuer and regulator consoles. Both demo moments working end to end. |
| <b>Weeks 1–6</b>            | Harden the pool: invariant tests, a re-run trusted setup ceremony with external contributors, and an external review of the circuits. Publish the threat model and the honest privacy scope.                                                             |
| <b>Months 2–4</b>           | Pilot with one tokenized-note issuer — ideally from the Founding Circle — that has a real concentration or jurisdiction limit. Success is one regulator-facing audit pack accepted on the strength of its proofs.                                        |
| <b>Months 4–9</b>           | Package as an attachable module with an SDK, so an issuer adds a confidential register without changing issuance. Extend to the chains Cleanverse already spans.                                                                                         |

**What would make me stop**

If issuers tell me a public register is not actually blocking their allocators, the premise is wrong and no amount of cryptography fixes it. The pilot is designed to find that out early rather than late.

# Solo builder. Zero-knowledge financial infrastructure, end to end.

Circuits, contracts, and the application on top — shipped repeatedly under hackathon constraints.

| BUILD                                           | WHAT IT PROVES ABOUT THE SCOPE HERE                                                                                                                                                                                                                 |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Privacy-pool payment corridor</b><br>Stellar | Seven Circom/Groth16/BN254 circuits and eight contracts live on testnet, including the four selective-disclosure proof types and the in-circuit compliance proof Saksi is built on. <a href="https://tukar-six.vercel.app">tukar-six.vercel.app</a> |
| <b>Sealed-bid OTC desk</b><br>Stellar           | Confidential order handling with settlement guarantees.                                                                                                                                                                                             |
| <b>Confidential OTC desk</b><br>Zama fhEVM      | The same problem under fully homomorphic encryption rather than zero-knowledge — two independent approaches to confidential markets.                                                                                                                |
| <b>Beyond ZK</b>                                | Shipped on Sui, on EVM chains, with EIP-7702 account abstraction, and on TEE-based systems.                                                                                                                                                         |